

Project Q2

Multi-Agent AI Framework for Industrial Fleet Management and Autonomous Troubleshooting

Background

AROL S.p.A. is a world leader in the design and production of advanced capping and filling machinery for bottles and jars, as well as capsule production. The company features high vertical integration: 95% of components and applications are developed in-house, ensuring total control over quality and production efficiency. With operational headquarters in Turin, other Italian cities, and key markets including the USA, Mexico, Brazil, China, and India, AROL provides a global presence serving various industrial sectors.

In recent years, AROL has undergone significant technological evolution, similar to that of the automotive sector. This shift has led to an increasing integration of advanced IT skills for centralized data management and machinery performance monitoring. AROL stands out for adopting advanced solutions across operating systems and system programming, web programming, modern user interfaces, machine learning, applied AI, and image recognition for quality control.

Project's Summary

As part of a broader Digital Business Transformation, AROL is developing a customer-dedicated platform to facilitate direct, immediate communication between the company and end users.

The goal is to provide customers with an advanced fleet management tool that lets them easily access all information about their machines: from the initial quote (including historical revisions) to orders, contractual documents, user manuals, and IoT data.

This platform serves as a digital archive collecting the "know-how" of the customer's machinery. Leveraging the vast amount of available data, AROL aims to integrate an agentic AI solution.

This AI will assist customers in retrieving information and providing operational support, such as technical troubleshooting.

By scanning a QR code on the machine, the end-user should quickly access the platform section dedicated to the user manual. Here, in addition to viewing the PDF, the user will be greeted by an AI Chatbot that guides them through the platform and, by invoking a series of AI Agents, provides all necessary support information.

The required activity involves developing the back-end and front-end of the chatbot, the AI agent architecture, and the relative orchestrator.

System Architecture

1. Frontend Interface: A responsive web dashboard (React/Angular/Vue) accessible via QR code, optimized for mobile industrial use.
2. Back-end Services: A microservices architecture (Node.js/Python) to manage user sessions and bridge the UI with the AI Orchestrator.
3. AI Orchestrator: A central "brain" (using frameworks like LangGraph or AutoGen) that interprets user intent and delegates tasks to specialized agents.

4. Specialized Agentic Layer:
 - o Doc-Agent: Uses RAG (Retrieval-Augmented Generation) to extract precise instructions from technical PDFs.
 - o Telemetry-Agent: Connects to IoT data streams to diagnose machine health status.
 - o Business Agent: Retrieves order history and maintenance contracts from the corporate database.

Objectives

The project aims to design and implement an intelligent ecosystem composed of a conversational interface (Chatbot) and a multi-agent AI system. The system must be capable of performing "reasoning" over heterogeneous data sources (PDF manuals, IoT telemetry, contractual history) to provide real-time expert assistance to plant operators.

Required Background

Knowledge of AI, Machine Learning, Neural Networks, Agents.

Working Environment

Students may work on a laptop or desktop, under Unix-like or Windows systems.
The project can be naturally extended into an MSc thesis.

Deliverables

All projects must be delivered including:

- Source files.
- A README text file (plain ASCII) with instructions on how to compile and run the program, execution parameters, and dataset formats.
- A DOCUMENTATION file (Word, LaTeX, Markdown, etc.) containing:
 - o main design choices (graph format, memory layout, CPU and GPU parallelization schemes, strategies for handling highly skewed degree distributions);
 - o an experimental evaluation with tables and plots comparing:
- A set of OVERHEAD slides (PowerPoint or similar) for a 20-minute oral presentation of the projects.

Contact

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Additional Information

Additional details will be provided during the visit to Canelli on May 20th, 2026. It is recommended that at least one member of each group working on this activity attends the visit.